

Abhishek Subedi

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Summary

Graduate Mechanical Engineering student (MS, May 2027) with hands-on experience in CAD design and CFD-driven thermal analysis. Proficient in ANSYS, STAR CCM+, SolidWorks, Python and MATLAB, having developed experimental flow setups and automated control logic for supersonic test facilities. Skilled in geometry cleaning, meshing and multiphysics simulations, seeking a Thermal Modeling Engineer internship to assess battery pack safety risks.

Education

University of Toledo

M.S., Mechanical Engineering

•GPA: 4.0/4.0

Aug 2024 - May 2027

Toledo, OH

Kathmandu University

B.E., Mechanical Engineering

•GPA: 3.5 / 4.0

Aug 2018 - May 2023

Dhulikhel, Nepal

Experience

University of Toledo

Graduate Research & Teaching Assistant

Jan 2025 - Present

Toledo, OH

- Experimentally characterizing drag coefficients of particles in particle-laden supersonic flow using Schlieren imaging and PIV/PTV; processing multi-modal optical datasets through MATLAB and Python pipelines from raw acquisition through drag-coefficient and velocity-field extraction, with iterative corrections improving measurement accuracy and reliability.
- Designed and deployed instrumentation, flow-conditioning hardware, and data-acquisition systems for the supersonic flow facility; developed control logic with safety interlocks and delivered a complete CAD drawing package and engineering documentation, enabling repeatable experimental campaigns.
- Guide Fluid Mechanics Laboratory students through experimental setups, equipment operation, and data analysis using Pandas for data processing, resulting in clearer understanding of core concepts and higher lab report scores
- Assist Thermodynamics course instruction by leading weekly office hours, grading assignments, and providing feedback, which helped students grasp concepts better and improve their exam performance
- Supervise Heat Transfer Laboratory experiments and enforce safety protocols, applying experimental testing methodologies and thermal modeling validation to maintain zero safety incidents and ensure consistent equipment performance throughout the semester

Mahavir Shree International Pvt. Ltd.

Associate Engineer

Apr 2024 - Aug 2024

Kathmandu, Nepal

- Led mechanical systems design, relocation, and installation of baggage handling systems at TIA Domestic Terminal using SolidWorks and AutoCAD for 3D modeling and engineering drawings; coordinated with vendors and managed upgrades to ensure on-time completion, improved reliability, and reduced waiting time for incoming passengers.
- Supervised scheduled maintenance of Rosenbauer Panther 6x6 fire vehicles, tracking non-conformances and coordinating corrective actions using the maintenance management system; ensured OEM compliance and updated technical documentation, which increased vehicle availability and reduced downtime

Sipradi Trading Pvt. Ltd.

Engineering Intern

Jan 2023 - Apr 2023

Lalitpur, Nepal

- Supported service and maintenance of TATA vehicles' engine, transmission, and EV systems by performing mechanical assembly, field verification, and BOM checks; applied knowledge from the formal OEM training program to diagnose issues and help reduce vehicle downtime
- Assisted in reorganizing workflow using NAV software and AutoCAD-based standardized assembly drawings; utilized MATLAB scripts to automate data processing, increasing documentation accuracy and cutting turnaround time in the high-volume service environment.

Projects

Automatic Pipe Cleaning Crawler (Custom Machine Design)

- Designed and fabricated a link-and-spring mechanism enabling adaptive radial contact for self-centering navigation within variable-diameter pipelines; modeled components in SolidWorks, selected parts, assembled prototype, and validated performance through mechanical testing. Documented design specifications and test results in technical reports.

Vehicle Chassis and Body Design: Shell Eco-Marathon (Team Junkiri)

- Led full SolidWorks chassis and surface modeling for competition vehicle; produced assembly drawings with ballooning, managed parts list, and coordinated the fabrication workflow for international competition in Indonesia.

Hydrogen Fuel Cell Powertrain (Retrofitting a Maruti 800)

- Specified and modeled integration of H₂ fuel cell components into an existing ICE chassis; performed system-level layout, packaging, and design feasibility analysis. Published in IOP Conference Series (2023).

Battery Thermal Management in EVs (Comparative Analysis)

- Evaluated active cooling strategies and equipment configurations for EV battery packs; performed thermal load analysis to inform design trade-off decisions.

Skills

- **CAD / Design:** SolidWorks, AutoCAD, 3D modeling, multi-component assemblies, assembly drawings and ballooning, BOM
- **Design Practice:** Design for manufacturability, component selection & supplier coordination, drawing packages, non-conformance resolution, technical documentation
- **Fabrication & Testing:** Prototype build, mechanical assembly, test setup, field verification, Experimental Testing
- **Programming:** Python, NumPy, Pandas, SciPy, Matplotlib, MATLAB, Git, GitHub
- **Simulation & Analysis:** Thermal Modeling, Computational Fluid Dynamics, Heat Transfer, Numerical Methods, Multi-physics Simulations, STAR CCM+, ANSYS

Publication

- Y. Shrestha, A. Subedi, M. Sapkota, P. Paudel, N. K. Yadav, and B. S. Thapa. Assessing the performance of a demonstrative hydrogen fuel cell power train in the chassis of an internal combustion engine vehicle. *IOP Conference Series: Materials Science and Engineering*, vol. 1279, no. 1, p. 012003, Mar. 2023. DOI: 10.1088/1757-899X/1279/1/012003.

Training & Certifications

- **SolidWorks Associate – Mechanical Design (CSWA):** Dassault Systèmes, Nov 2022
- **Capacity Development Program on EV Technology:** Kathmandu University, May 2023. EV components, power electronics, charging systems, design/simulation.
- **Data Science with Python:** Broadway Infosys, Kathmandu, Nepal, Apr – Jul 2024. Python, NumPy, Pandas, SciPy; data visualization; Git; ML fundamentals.